TASK 1: FILE HANDLING

Python Programming Practical Report — 20 Marks

1. Aim

To create a text file, write data into it, read and display the data, append new data to the existing file, and display the updated file content using Python file-handling operations.

2. Problem Statement

Write a Python program that demonstrates the basic file operations: creating a file, writing data, reading data, appending new data, displaying the updated content, and closing the file after each operation.

3. Concepts Used

Concept	Purpose
File opening	Opens a file in the required mode.
Writing (w)	Creates a file or overwrites its existing content.
Reading (r)	Reads and displays the file content.
Appending (a)	Adds new data to the end of the existing file.
Closing	Closes the file after the operation.

4. Source Code

```
# Task 1: File Handling

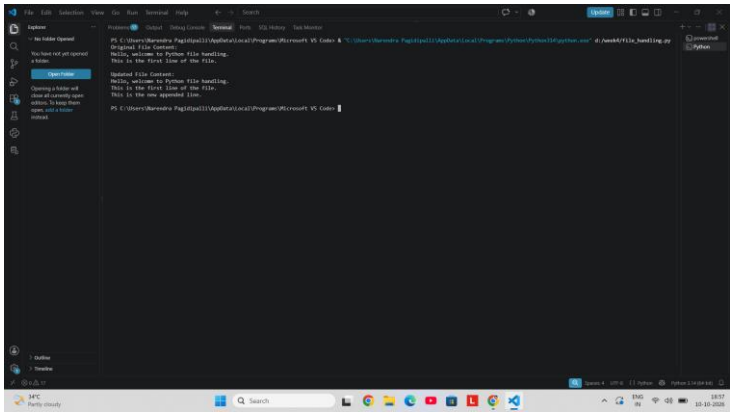
# Create a file and write data
file = open("sample.txt", "w")
file.write("Hello, welcome to Python file handling.\\n")
file.write("This is the first line of the file.\\n")
file.close()

# Read and display the data
file = open("sample.txt", "r")
print("Original File Content:")
print(file.read())
file.close()

# Append new data to the file
file = open("sample.txt", "a")
file.write("This is the new appended line.\\n")
file.close()

# Display the updated file content
file = open("sample.txt", "r")
print("Updated File Content:")
print(file.read())
file.close()
```

5. Output



6. Project Explanation

1. The "w" mode creates the file and writes the first two lines. If the file already exists, its previous content is overwritten.
2. The "r" mode reads the file content and displays it on the screen.
3. The "a" mode appends a new line without deleting the existing content.
4. The file is opened again in "r" mode to display the updated content.
5. The close() method closes the file after each operation.

7. Challenges Faced

Understanding the difference between write, read, and append modes can be challenging. This is addressed by opening the file separately in each mode and checking the output after each operation.

8. Conclusion / Result

The program successfully creates a text file, writes data, reads and displays the content, appends additional data, and displays the updated file content. Thus, the basic file-handling operations in Python are demonstrated.